ISSN 1069-3513, Izvestiya, Physics of the Solid Earth, 2022, Vol. 58, No. 2, pp. 185–202. © Pleiades Publishing, Ltd., 2022.
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Amplitude of Secular Geomagnetic Variation in Late Cretaceous Based on Paleomagnetic Studies of the Okhotsk–Chukotka Volcanic Belt from Upper Reaches of Malyi Anyui River, West Chukotka

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Received May 18, 2021; revised June 20, 2021; accepted June 20, 2021

Abstract—A paleomagnetic study of the Okhotsk–Chukotka belt volcanics exposed in the region of the Kupol field (~66.9° N, 170.1° E) has been carried out. The studied rocks were formed between 85 and 95 Ma during the Cretaceous Normal Superchron (CNS). In most of the studied samples, the characteristic magnetization component is isolated. Its primary origin is supported by the positive fold test, magnetic mineralogy data, and similarity of the calculated paleomagnetic pole with that of coeval rocks from the same region. It is shown that the studied rocks were formed in high polar latitudes close to 80° N. The amplitude of paleosecular variation (PSV) of the geomagnetic field during CNS has been estimated. Depending on the calculation method, the estimates (and their confidence intervals) for ~81° N are $Sb = 22.1°$ (19.5°; 24.7°) in case of using a fixed cut-off angle of 45° and $Sb = 23.1°$ (20.3°; 25.7°) in case of the cut-off angle calculated by the Vandamme method. The obtained estimates generally support the existing models of PSV latitudinal dependence for CNS. At the same time, the estimates are less consistent with the model (McFadden et al., 1991) where PSV values are low at the equator and sharply increase with latitude than with the model (Biggin et al., 2008) predicting higher PSV values at the equator and their more gentle increase towards polar latitudes. Irrespective of the calculation method, the obtained estimates almost perfectly fit the PSV latitudinal dependences for the last 5 Ma. This means that the amplitude of secular geomagnetic variation during CNS, at least at high latitudes, was the same as it was during the last 5 Ma. The obtained result does not support the hypothesis that geodynamo regimes change at the transition from the periods of frequent geomagnetic reversals to superchrons (e.g., (Gallet and Pavlov, 2016)), which suggests, in particular, that the modern numerical models are not fully adequate to the real processes of generation of the geomagnetic field.

Keywords: Late Cretaceous, amplitude of secular variation, Okhotsk–Chukotka volcanic belt

DOI: 10.1134/S1069351322020045

INTRODUCTION

The dynamics of changes in the polarity of the geomagnetic field during the last 150 Ma is well known from the study of the marine magnetic anomalies and due to extensive magnetostratigraphic research. For the more ancient periods of the Phanerozoic, the magnetostratigraphic data obtained from the well biostratigraphically dated sections are not as ample and they provide an incomplete and sometimes fragmentary record of the evolution of the geomagnetic field. Nevertheless, the existing information confidently indicates that the frequency of geomagnetic reversal substantially varied during the Phanerozoic from zero during the so called superchrons to rather high values (7–8 over Ma) in the Miocene, Mid-Jurassic, and Mid-Cambrian (e.g., (Pavlov and Gallet, 2020; Gallet et al., 2012; Opdyke and Channell, 1996). To date, three Phanerozoic superchrons have been established, each with a duration of at least several tens of millions of years: the Ordovician Moyero (Pavlov and Gallet, 2005), the Late Paleozoic Kiama (e.g., (Opdyke and Channell, 1996)), and the youngest one—the Cretaceous superchron discovered by Helsley and Steiner (1968). Several superchrons have been also identified the Precambrian (e.g., (Driscoll, 2016)), among which the two most confidently established are the Meso-Neoproterozoic Maya superchron (Gallet et al., 2012; Pavlov et al., 2019) and the Early Mesoproterozoic Belt superchron (Elston et al., 2002).

Why the magnetic reversals typically quite frequently (on the geological time scale) freezes up for tens of millions of years in one of its two possible sta-

tionary states is still unclear. Most authors relate the occurrence of superchrons to the long-period variation (a characteristic time of the order of 10 Ma and longer) in the conditions at the core–mantle boundary (CMB) (e.g., (McFadden et al., 1991)), which, in turn, can be caused by various mantle processes. Importantly, the numerical geodynamo simulations (e.g., (Coe and Glatzmaier, 2006; Landeau et al., 2017)) support these views.

On the other hand, Hulot and Gallet (2003) note that the transitions between the reversing and nonreversing (superchron) geodynamo states can be sudden, perhaps indicating that the occurrence of superchrons can probably be a consequence of the nonlinear nature of the magnetohydrodynamic processes in the Earth's core and can be purely stochastic. If this is confirmed, the very idea of the connection between the changes in the frequency of geomagnetic reversals and the processes in the mantle and at CMB will be challenged. Since the first standpoint implies a necessary substantial change in the boundary conditions for the superchrons to occur, then, if this standpoint is correct, the geomagnetic field parameters during the superchrons should significantly differ from those for the periods of the normally reversing field. On the contrary, if Hulot and Gallet (2003) are right, the transition to the superchron state is not necessarily connected with changes in the geomagnetic field parameters.

Practically all numerical models proposed to date cannot produce superchrons without changing the boundary conditions. On the one hand, this can be interpreted in support of the first standpoint; on the other hand, it can testify to the inadequacy of the models. The latter is likely to be supported by the recent results (Shcherbakov and Fabian, 2012) that the known durations of the geomagnetic polarity chrons can be described by a distribution in which superchrons will not be anomalous events. In the latter case, we may expect the geomagnetic field parameters of a superchron to not differ substantially from those in the periods of frequent reversals. Solving the above questions requires sufficiently large amount of high-quality data describing the behavior of the field within and beyond the superchrons.

The main parameters of the geomagnetic field that can be used for comparing its states within and beyond superchrons are field geometry (i.e., the ratio of dipole and nondipole components), field strength, and character of paleosecular variations (PSV). Studying the geometry of the ancient magnetic field is an extremely challenging task which requires huge data distributed over different latitudes for its strict solution. In this work, we do not consider this issue.

Thus, we only have a field strength and its secular (paleosecular) variation to study superchrons. We plan to present the data on the field strength in our next paper. This work addresses the aspect of the problem that concerns secular variation of the directions of the geomagnetic field.

Rather, the data acquired in recent decades support the conclusion that the geomagnetic field during superchrons was much more stable than in the last 5 Ma (Biggin et al., 2008). This interpretation, however, is based on the determinations obtained from the very few studied sections. Moreover, practically all the existing determinations for the Cretaceous superchron were obtained from the low or middle latitudes, and only one estimate of the PSV amplitude for polar latitudes has been obtained so far (Tarduno et al., 2002). This significantly complicates the construction of the PSV latitudinal dependence which is currently the main characteristic allowing the study of the character of PSV changes during the geological time.

The recent analysis of a much larger data set which includes the results obtained in the last years on the amplitude of secular variation gives reason to doubt the validity of the claimed greater stability of the geomagnetic field during superchrons. However, the reliability of this analysis is also strongly limited by the absence of data on the high-latitude secular geomagnetic variation in the Cretaceous superchron. The study results from the rocks of the Okhotsk–Chukotka volcanic belt (OCVB), which are presented in this work, to an extent fill this gap.

A considerable part of the volcanics of the Okhotsk–Chukotka belt was formed during the Cretaceous Normal Superchron below 70°–80° latitudes of the northern hemisphere (Akinin and Miller, 2011; Tikhomirov, 2020; Stone et al., 2009) and is therefore an extremely important object for paleomagnetic examination to study the Late Cretaceous high-latitude variations. We carried out these studies in 2019–2021 on a few segments of the Okhotsk–Chukotka volcanic belt.

STUDY OBJECT AND SAMPLING

The Okhotsk-Chukotka volcanic belt stretches for more than 3000 km along the Pacific margin of Asia and is one of the Earth's largest subduction-related igneous provinces (Belyi, 1977; Tikhomirov, 2020; Akinin and Miller, 2011). Most researchers use the OCVB zoning scheme proposed by V.F. Belyi (1977). In this scheme, in accordance with the differences in the basement structures of the belt, six segments (or sectors, according to Belyi) are distinguished in OCVB from the southwest to the northeast, namely, the West Okhotsk, Okhotsk, Penzhina, Anadyr, Central Chukotka, and East Chukotka segments.

Generally, the belt's volcanics do not have signs of significant deformations (Sokolov et al., 2010; Akinin et al., 2020; Tikhomirov, 2020). The attitude of the volcanic units is typically controlled by the primary relief and by structures of compensating subsidence—

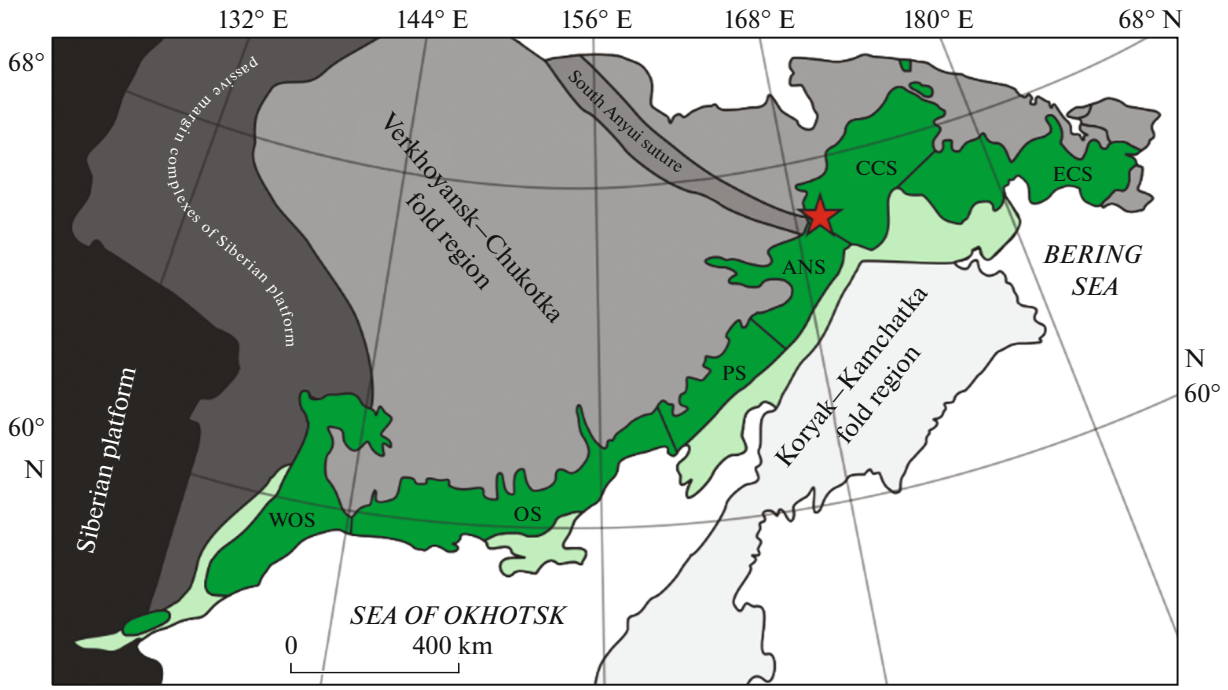


Fig. 1. Tectonic scheme of northeastern Asia according to (Tikhomirov, 2020; Sokolov, 2010) with simplifications. Light green is Late Jurassic–Early Cretaceous (Pre-Albian) volcanic complexes of Pacific active margin; green is Okhotsk–Chukotka volcanic belt with segment boundaries: ECS, East Chukotka segment; CCS, Central Chukotka segment; ANS, Anadyr segment; PS, Penzhina segment; OS, Okhotsk segment; WOS, West Okhotsk segment; red asterisk is study object.

calderas and volcano-tectonic depressions (Tikhomirov, 2020).

OCVB was formed beginning from 106 and ending 76 Ma (Akinin et al., 2020, with the most powerful volcanic pulses between 89 and 80 Ma (Tikhomirov, 2020; Akinin and Miller, 2011). Thus, a significant part of the OCVB volcanics was formed during the Cretaceous Normal superchron.

The study region is located in the western part of Chukotka at the boundary of the Anadyr and Central Chukotka OCVB segments, to the northeast of the Kupol gold deposit, in the upper reaches of the Malyi Anyui River (Figs. 1 and 2).

Overall, nine sections were sampled in this region (Fig. 2). Each section includes from 4 to 17 sites, and each site has from 8 to 18 individually oriented samples. Each site is an individual lava flow, less frequently an outcrop of tuffs or ignimbrites. The samples were oriented using a magnetic compass with declination correction based on the 12th generation IGRF model (Thébault et al., 2015). During sampling, we controlled whether the sampled rocks affect the magnetic compass arrow.

The studied sections pertain to the Pykarvaam, Koekvun, and Emuneret formations (Malysheva et al., 2012). According to (*Resheniya*..., 2009) primarily based on gthe paleobotanical data, the first two formations are dated to the Albian (113.2–100.5 Ma (Gradstein et al., 2020)) and the latter one to the Turonian and Coniacian ages (93.9–85.7 Ma (Gradstein et al., 2020)). The Pykarvaam Formation is mainly composed of lavas, ignimbrites, and tuffs of acid composition; the Koekvun Formation, of andesitic and basalt lavas and tuffs of the same composition; and the Emuneret Formation, of acidic tuffs and ignimbrites (Malysheva et al., 2012). Most sections studied in this work pertain to the Albian Koekvun Formation.

The published $^{40}Ar/^{39}Ar$ and U-Pb and U-Pb isotopic dates closest to the sections are obtained for the samples from the vicinity of the Kupol deposit and upper reaches of the Ugatkyn River (Fig. 2). The $^{40}Ar/^{39}Ar$ ages of monofractions extracted from rhyolitic tuff of the Pykarvaam Formation at Mt. Kovalenko (Fig. 2) are 93.7 ± 1.2 Ma for amphibole and 91.8 ± 1.6 Ma for plagioclase (Tikhomirov et al., 2006). The $^{40}Ar/^{39}Ar$ age obtained for plagioclase from olivine basalts attributed to the Emuneret Formation is 86.8 ± 2.7 Ma (Tikhomirov et al., 2006). These dates characterize the upper part of the K section shown in Fig. 2. Zircon from rhyodacite tuff of the Pyakarvaam Formation from the upper reaches of the Ugatkyn River has an U-Pb age of 85.4±1.2 Ma (Tikhomirov, 2020). In the overall volcanics section present at this site, this sample is acquired somewhat above the uppermost point of section U (Fig. 2). The datings obtained from the vicinity of the Kupol deposit are ~3 km away from our extreme sections (1, 2, K), and a distance

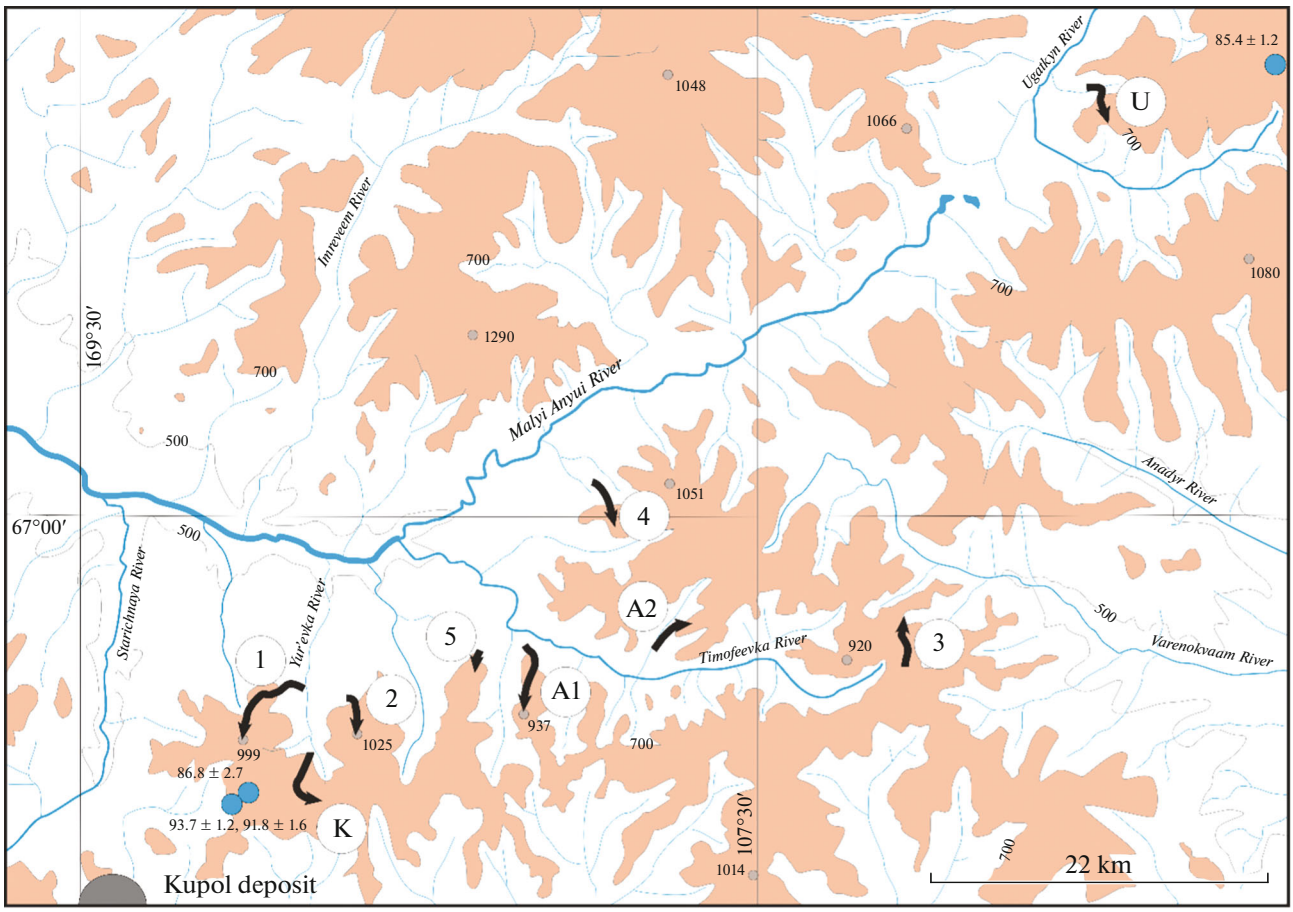


Fig. 2. Schematic layout of studied sections in region of Kupol deposit. Sections: U, Ugatkyn; K, Kupol, Mt. Kovalenko; A1, Anyuy 1, Mt. Kayemnay; A2, Anyuy 2, Mt. Timofeevka; 1, Mt. Belaya; 2, Yur'evka River; 3, upper reaches of Timofeevka River; 4, Anyuy River; 5, Vetvistyi creek; blue circles are sampling sites of closest isotope datings discussed in text and their ages.

between section U and the sample taken from the upper reaches of the Ugatkyin River is 13 km.

Tikhomirov (2020) argues in favor of the OCVB stratigraphic models based on the results of $^{40}Ar/^{39}Ar$ and U-Pb dating. Considering the lower reliability of paleofloristic data compared to isotopic data, we prefer datings obtained by isotopic methods. Due to the complexity of lateral correlations in the OCVB volcanic complexes, we estimate the age of our sections at 95–85 Ma.

Most of the studied sections except for the westernmost ones (1, 2, and K) are located within the gentle Ugatkyn (Ugatkyn-Yurumkuveem) monocline (Tikhomirov, 2020; Belyi, 1977). The monocline dips southeast at up to 15° and its total length reaches 120 km. In the study region is also a lower order structure—the Timofeevskaya subsidence (Senotrusov, 1990). This is a rounded in plan, gentle negative structure with a diameter of about 10 km, with the youngest volcanics outcropping in the center. The center of the structure is located in the upper reaches of the Timofeevka River. The northwestern limb of the subsidence is located along the strike of the Ugatkyn monocline. The morphology of the Timofeevskaya subsidence is quite similar to that of the downsag caldera type (Cole et al., 2005). The nature of the Ugatkyn monocline is less obvious. It was hypothesized that this structure is related to weak extension transverse to the volcanic belt strike (Tikhomirov, 2020). The idea that both structures emerged after the formation of the volcanics rather than reflect the paleorelief is favored, in particular, by the substantial size of the Ugatkyn monocline and the positive paleomagnetic fold test (details are in the next section).

Sections 1, 2, and K are located to the west of the monocline in the upper reaches of the Yurievka River, a left tributary of Malyi Anyi. The stratified volcanics in these sections have a horizontal attitude or very gently (1°–3°) dip south. All the three formations outcropping in this region have been sampled in these three sections. The rocks are mainly basic lavas—porphyritic, aphyric basaltic lavas, sometimes with olivine. Among 41 sites sampled in these sections, only five more acidic flows were diagnosed.

Sections A1, A2, 4, and 5 are located east of the above sections, in the lower reaches of the Timofeevka River. The sections capture the Pykarvaam and Koekvun formations. The rocks in these sections dip southeast at 5°–12°. The sections are mainly represented by basaltic and andesite basaltic lavas but acidic varieties are more common. Ignimbrites, dacite lavas, and rhyolite tuffs occurred in different parts of the section.

Sections 3 and U are located some distance away from the above sections (Fig. 2). Section 3 is located in the upper reaches of the Timofeevka River. Its position on the eastern limb of the Timofeevskaya subsidence makes this section particularly interesting for conducting the field paleomagnetic fold test. The strata in this section have a reversed westward dip but the dip angle is relatively small, only 3°. The section is composed of porphyritic basaltic lavas of the Koekvun Formation. Section U is located in the upper reaches of the Ugatkyn River, about 30 km northeast of the other sections. The studied section only encompasses the Pykarvaam Formation. In contrast to the above sections, the rocks composing section U predominantly have intermediate and acidic composition. The section begins from rhyolite/dacite lavas and their tuffs which also crown the section; lavas and tuffs of andesites and basalts are found in the middle of the section.

Confident correlation of the sections with each other is challenging because of the absence of any marker beds and the presence of significant lateral variations of the volcanic complexes. Based on the results of satellite image interpretation and field observations, in this study we assume (with possible partial overlapping) the following age sequence of the studied sections (from bottom to top): U, 4, 1, 2, K, 5, A1, A2, and 3.

PALEOMAGNETISM

The laboratory paleomagnetic studies were conducted by the standard technique (Khramov et al., 1982; Butler, 1998; Tauxe, 2010) in the paleomagnetic laboratories of Schmidt Institute of Physics of the Earth of the Russian Academy of Sciences (IPE RAS), Moscow, and Ludwig-Maximilian University, Munich. The electron-microscopic and microprobe studies were carried out on the TESCAN Vega II scanning electron microscope in IPE RAS Geophysical Observatory in Borok and on the Jeol JSM-6480LV electron microscope in the laboratory of Petrology Department, Faculty of Geology of Moscow State University.

At the first step of the laboratory study, we compared the results of thermal demagnetization and demagnetization by the alternating magnetic field (AF) on duplicate specimens of several samples from a number of sections. The comparison has shown that both demagnetization types give statistically indistinguishable results. In the subsequent experiments, we used thermal demagnetization for sections K and U and AF demagnetization for the remaining sections.

The samples from 74 sites out of the 79 studied ones contain a good and satisfactory paleomagnetic record. Their natural remanent magnetization (NRM) contains predominantly a single characteristic component which is stable to demagnetization.

Sometimes, samples also contain a low-coercive or low-temperature components which have chaotic directions and are most likely to be related to the present-day viscous remagnetization during the storage or transportation of samples. The characteristic high-coercive or high-temperature component has very steep inclinations and different declinations (Fig. 3 and Table 1).

The primary origin of the identified characteristic component is suggested by the positive fold test (field test), electron microscopy data and microprobe analysis, and coincidence of the calculated pole with the coeval poles of Chukotka obtained in the previous works (Stone et al., 2009; Lebedev et al., 2021).

For the fold test we used the Enkin's program package which implements two versions of the test (Watson and Enkin, 1993) and (Enkin, 2003). The test was performed at the section-mean level, with the rocks within each section having the same attitude in all the studied sites. Comparing mean directions at the section rather than site level avoids excessive influence on the results by the sections with more sites than others. The Watson–Enkin test (Watson and Enkin, 1993) yields a maximum at 97.8% of unfolding with a 95% confidence interval of 63.6–133%. Accordiong to the Enkin test (Enkin, 2003), an untilting value (direction-correction (DC) parameter) is 0.98 ± 0.63. Thus, both tests are positive indicating that the magnetization components isolated in the studied rocks were firmed before tectonic deformation (tilting). The relatively large confidence intervals are due to the small difference in attitude elements between the studied sections.

Since the deformations of the primary attitude of rocks on the studied segments are likely to have occurred during compensatory subsidence immediately following the formation of the volcanic sections, positive test definitely indicates that the isolated characteristic magnetization component was formed during or shortly after the effusion of lava units.

The electron microscopic studies and microprobe analysis of volcanic samples revealed titanomagnetite grains with structures of high-temperature heterophase oxidation with ilmenite lamellas and low-Ti titanomagnetite between them (Fig. 4, samples 010108A, B, 020407). These structures are formed directly during cooling of rocks after their crystallization from magma and definitely indicate the primary magmatic origin of the respective grains.

Besides, the studied basalts are abundant in primary magmatic small homogeneous idiomorphic tita-

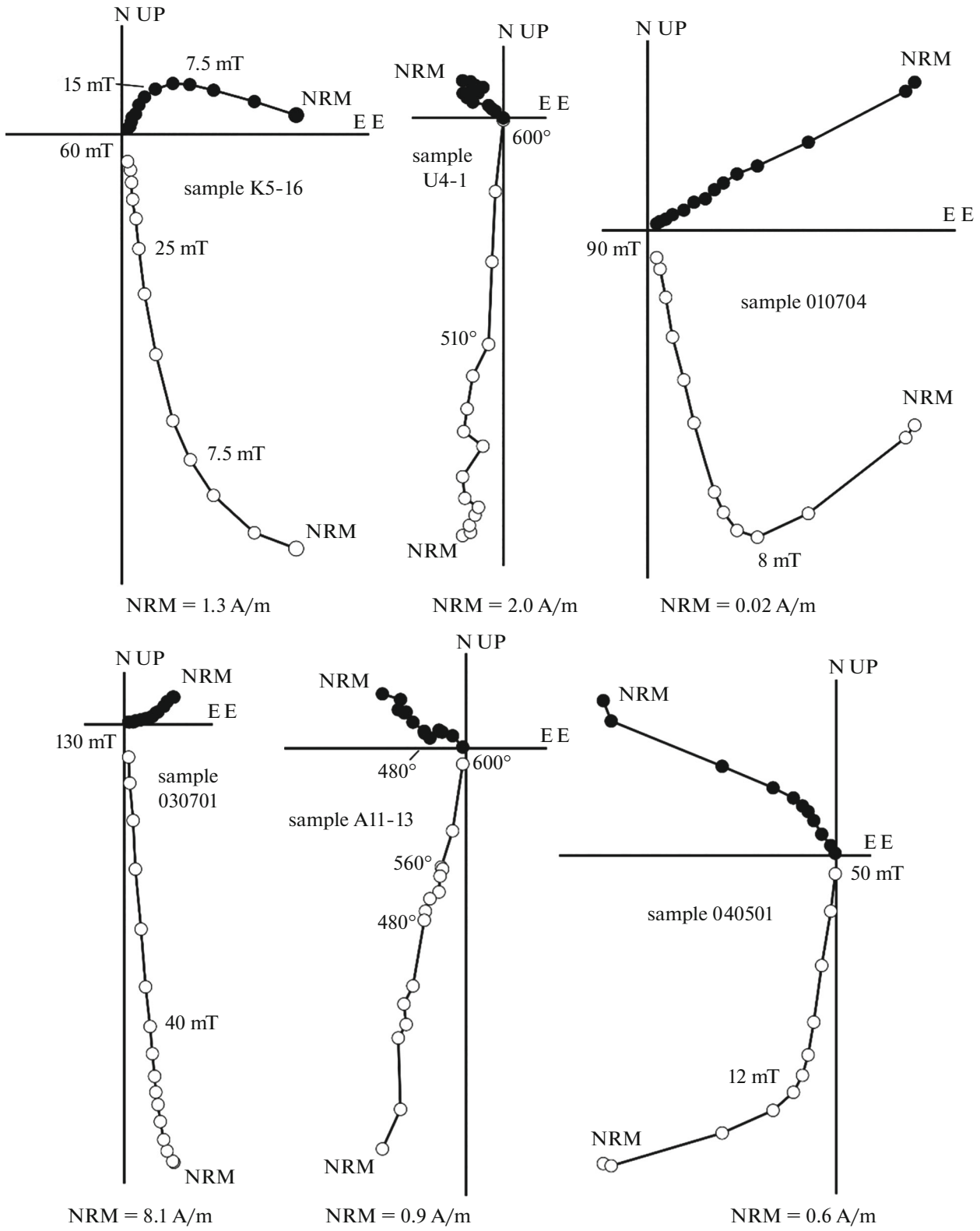


Fig. 3. Examples of Zijderveld diagrams based on thermal demagnetization results for rocks from sections K (sample K5-16), U (sample U4-1), 01 (sample 010704), 03 (sample 030701), A1 (sample A11-13), and 04 (sample 040501). Black (white) circles are projections of NRM vector on horizontal (vertical) plane.

Table 1. Paleomagnetic directions determined in studied rocks

Site no.	Formations and rock composition data according to (Malysheva et al., 2012)	*N*	*n*	Dg	Ig	Ds	Is	K	a95	VGP for individual sites			Km, $\times 10^{-3}$ Si units	VGPs for site groups		
										PLONG	PLAT	A95		PLONG	PLAT	A95
U (Ugatkyn)																
67.2° N, 171° E; dip azimuth 140°, dip angle 4°																
U07	Pykarvaam Formation: ignimbrites, rhyolite and dacite tuffs, dacite vitrophyres	14	13	279.1	75.5	266.5	78.2	93.9	4.3	125.6	57.3	8.2	12.8			
U06		13	10	342.9	80	356.6	83.5	89.5	5.1	166.7	80.0	10	7.09			
U05		13	11	257.7	72	245.7	73.5	87.8	4.9	129.6	45.4	8.5	6.35			
U04		15	14	329.6	86.2	71.8	89.3	104	3.9	174.5	67.6	7.4	22.7			
U03		12	12	247.7	84.2	208.1	84.0	80.8	4.9	160.9	56.3	9.1	12.7			
Mean		**8**	**5**	**282.3**	**81.4**	**258.3**	**84**	**85**	**8.3**	**145.3**	**62.4**	**16.1**				
NRO											0.087				0.087	
04 (Anyui)																
67° N, 170.3° E; dip azimuth 149°, dip angle 6°																
0401	Pykarvaam Formation: ignimbrites, rhyolite and dacite tuffs, dacite vitrophyres	12	11	269.3	80.1	232.6	81.4	644	1.8	147.1	54.3	3.4	11.7			
0405		15	9	1.9	73.9	18	78.5	159	4.1	245	83.1	7.5	1.48	281.7	84.7	3.2
0403	Koekvun Formation: basalts, andesite-basalts, andesites, and their tuffs	15	14	5	77.7	30	81.8	299	2.3	212.1	78	4.4	12.6			
0404		15	13	14.1	74.6	34.5	78.1	159	3.3	243.7	76.7	6	9.34			
Mean		**4**	**4**	**353.9**	**79.1**	**18.5**	**84**	**72.4**	**10.9**	**187.6**	**77.6**	**21.4**				
NRO											0.71				0.5	
1 (Mt. Belaya)																
66.9° N, 169.8° E; dip azimuth 170°, dip angle 3°																
0107	Pykarvaam Formation: ignimbrites, rhyolite and dacite tuffs, dacite vitrophyres	14	9	44.2	78.6	58.2	80.1	77.1	5.9	222.8	69.5	11.1	1.06			
0101		13	8	216.8	78.7	207.6	76.5	86.2	6	154	42.7	10.7	6.95			
0112	Koekvun Formation: basalts, andesite-basalts, andesites, and their tuffs	14	10	37.6	75.1	47.2	76.9	54.5	6.6	243.3	71.2	11.9	23.6			
0113		12	6	305	69.4	298.7	71.4	204	4.7	88.9	60.3	7.7	24.8			
0104		15	12	286.3	70.3	278.3	71.4	98.2	4.4	104.3	52.6	7.2	5.17	101.1	56.5	4.7
0105		8	7	295.4	73	286.6	74.5	109	5.8	104.7	59.2	10	35.4			
0108		14	9	355.5	59.9	356.1	62.9	107	5	357.1	67.3	7	25.5			
0109		10	10	10.1	71.1	13.6	73.9	244	3.1	302.2	80.8	5.3	22.8			
0110		12	9	346.2	72	345.5	75	144	4.3	47.5	82	7.5	21.9			
0111		13	10	42.1	81.2	60.8	82.7	171	3.7	208.7	69.8	7.1	26.5			
0114		12	11	24.3	74.3	31.4	76.7	181	3.4	254.9	77.1	6.1	24.4			
0115		10	9	53.1	75.6	64.5	76.7	151	4.2	234.1	64.7	7.5	35	225.2	66.8	6.7
0116		15	8	46.8	80.6	64.6	81.9	78.2	6.3	212.4	68.6	12.1	38.3			

Table 1. (Contd.)

Site no.	Formations and rock composition data according to (Malysheva et al., 2012)	*N*	*n*	Dg	Ig	Ds	Is	K	a95	VGP for individual sites			Km, $\times 10^{-3}$ Si units	VGPs for site groups		
										PLONG	PLAT	A95		PLONG	PLAT	A95
0117	Emuneret Formation: ignimbrites, rhyolite and dacite tuffs	10	7	274.9	80.1	257.4	80.4	240	3.9	134.1	57.6	7.3	9.28			
Mean		**17**	**14**	**350.5**	**79.9**	**350.7**	**82.9**	**34.9**	**6.8**	**159.1**	**80.1**	**12.8**				
NRO											0.9499				0.4445	
						2 (Mt. Yur'evka)										
						66.9° N, 169.9° E; dip azimuth 0°, dip angle 0°										
0201	Koekvun Formation: basalts, andesite-basalts, andesites, and their tuffs	11	10	207.2	84.3	207.2	84.3	298	2.8	160.6	56.5	5.5	16.7			
0202		12	11	33.5	65.1	33.5	65.1	395	2.3	291.9	63.7	3.3	21.6			
0203		16	13	21	72.2	21	72.2	180	3.1	294.6	76.4	5.2	21.4	297.6	81.8	3.5
0206		11	10	8.8	76.3	8.8	76.3	321	2.7	294.2	85.3	4.8	38.1			
0205		11	8	4.5	75.8	4.5	75.8	129	4.9	320.9	85.8	8.7	30.5			
0204		11	9	352.5	78.3	352.5	78.3	159	4.1	95	87	7.5	39.9			
0210		12	10	2.8	85.7	2.8	85.7	244	3.1	171.6	75.4	6.1	7.66			
0209		11	11	119.4	66.2	119.4	66.2	644	1.8	214.1	34.2	2.7	9.37			
0207		12	8	167.2	83.8	167.2	83.8	183	4.1	174.6	54.8	8	16.2			
0208		12	10	36.6	68.1	36.6	68.1	163	3.8	282.7	66.1	5.9	5.77			
0211		12	12	69.7	73	69.7	73	279	2.6	240.9	58.8	4.4	15.8	239.5	62.1	3.8
0212		11	9	59.5	76.2	59.5	76.2	195	3.7	238.8	66	6.6	14.5			
0213		12	7	122.2	73.1	122.2	73.1	131	5.3	206.6	42.6	8.9	16			
0214		11	12	72.5	82.9	72.5	82.9	185	3.2	206.2	67.1	6.2	15			
0215		14	12	350.8	77.5	350.8	77.5	119	4	71.7	86.2	7.2	2.11			
0216		12	11	285.8	88.1	285.8	88.1	153	3.7	160.3	67.6	7.4	7.32			
Mean		**16**	**16**	**46.3**	**81.3**	**46.3**	**81.3**	**41.6**	**5.8**	**215.1**	**73.7**	**10.7**				
NRO											0.9519				0.676	
						K (Kupol, Mt. Kovalenko)										
						66.8° N, 169.8° E; dip azimuth 0°, dip angle 0°										
K08	Koekvun Formation: basalts, andesite-basalts, andesites, and their tuffs	14	9	316.2	67.8	316.2	67.8	167	4	65.2	63.1	6.1	22.3			
K07		12	13	356.1	72	356.1	72	73.3	4.9	2.1	80	8.1	23			
K06		14	13	85	85.2	85	85.2	77.6	4.7	193.5	65.8	9.3	19.6			
K05		16	14	327.2	84.8	327.2	84.8	97.9	4	148.5	74.5	7.9	23.2			

Table 1. (Contd.)

Site no.	Formations and rock composition data according to (Malysheva et al., 2012)	*N*	*n*	Dg	Ig	Ds	Is	K	a95	VGP for individual sites			Km, $\times 10^{-3}$ Si units	VGPs for site groups		
										PLONG	PLAT	A95		PLONG	PLAT	A95
K04	Emuneret Formation: ignimbrites, rhyolite and dacite tuffs and lavas	13	7	34.2	64.5	34.2	64.5	275	3.6	291.5	62.9	5.2	15			
K03		13	9	9.3	79.9	9.3	79.9	153	4.2	208.9	85.1	7.9	3.17	233	82.6	4.5
K02		13	10	26	78.3	26	78.3	302	2.8	243.9	80	5.1	5.37			
K01		14	11	47.1	83.5	47.1	83.5	544	2	203.4	72.9	3.9	5.29			
Mean		**8**	**8**	**7.6**	**79.3**	**7.6**	**79.3**	**52.6**	**7.7**	**208.2**	**85.6**	**13.6**				
NRO											0.7888				0.7441	
5 (Vetvistyi)																
66.9° N, 170.1° E; dip azimuth 107°, dip angle 5°																
0504	Koekvun Formation: basalts, andesite-basalts, andesites, and their tuffs	15	9	337.9	77.7	0.6	80.1	72.2	6.1	173	86.1	11.4	16.6	132.3	84	7
0503		14	10	320.8	76.5	337.2	80.3	102	4.8	117.8	80.9	9	14.5			
0501		18	14	241.7	84.2	184.5	85.8	155	3.2	168.8	58.6	6.3	14			
Mean		**3**	**3**	**316.2**	**81.2**	**344.8**	**84.9**	**98.3**	**12.5**	**159.1**	**76.2**	**26.1**				
NRO											0.8334				0.5	
A1 (Anyui 1, Mt. Kaiemnai)																
66.9° N, 170.2° E; dip azimuth 146°, dip angle 6°																
a20	Koekvun Formation: basalts, andesite-basalts, andesites, and their tuffs	16	11	5.1	78	32.1	81.8	174	3.5	212.6	77.4	6.7	19			
a19		13	13	341.2	76.7	352.9	82.3	127	3.7	157.3	81.7	7.1	17.6			
a18		14	12	258	79.4	224.7	80	87.2	4.7	148.5	50.8	8.8	34.2	143.6	53.8	6.2
a17		12	8	283.4	77.2	257.9	80.7	144	4.6	135.1	58.1	8.7	18.1			
a15		10	6	251.5	82.4	207.8	81.7	174	5.1	158	51.8	9.7	17.8			
a13		15	15	340.8	57.8	343.8	63.6	160	3	20	66.7	4.2	20			
a11		13	13	321.6	62.7	320.4	68.7	196	3	62.3	65.7	4.7	19.4			
a12		14	12	333.4	76.1	338.9	82	102	4.3	136.4	79.9	8.2	13.1			
Mean		**8**	**8**	**322.2**	**76.2**	**319.3**	**82.2**	**41.6**	**8.7**	**132.5**	**74.3**	**15.8**				
NRO											0.9927				0.8312	
A2 (Anyui 2, Timofeevka)																
66.9° N,170.4° E; dip azimuth 103°, dip angle 12°																
a04	Koekvun Formation: basalts, andesite-basalts, andesites, and their tuffs	13	11	332.1	70.2	8.2	75.1	226	3	310.9	84	20.8	16.3	359.2	88.6	3.4
a05		15	14	328	70.8	5.4	76.4	98.8	4	308.5	86.5	21.9	28.6			
a06		13	13	316.9	67.9	339.7	77.9	82.6	4.6	89	82.1	18.2	18.1			
a07		13	13	317	73.2	0.7	80.5	242	2.7	173.2	85.4	7.9	23.5			
a09		13	13	294.4	76.1	342.3	86.8	386	2.1	163.8	72.9	11.9	20.1			
a10		13	10	10.3	80.7	63.8	75.2	44.4	7.3	239.9	63.4	18.2	24.3			
Mean		**6**	**6**	**323.6**	**74**	**11.2**	**80**	**119**	**6.2**	**210.8**	**84.5**	**11.5**				
NRO											0.8727				0.8364	

Table 1. (Contd.)

Site no.	Formations and rock composition data according to (Malysheva et al., 2012)	*N*	*n*	Dg	Ig	Ds	Is	K	a95	VGP for individual sites			Km, $\times 10^{-3}$ Si units	VGPs for site groups		
										PLONG	PLAT	A95		PLONG	PLAT	A95
3 (upper reaches of Timofeevka River)																
66.9° N, 170.7° E; dip azimuth 239°, dip angle 3°																
0302	Koekvun Formation: basalts, andesite-basalts, andesites, and their tuffs	11	11	76.8	86.4	127.7	88.8	249	2.9	175.3	65.4	5.8	48.5	168.5	71	4.4
0303		12	10	23.8	85	349.7	86.9	321	2.7	166.9	72.9	5.4	45.5			
0304		11	10	10.3	84.4	338.5	85.7	72.8	5.7	158.9	74.6	11.3	48.1			
0307		12	9	129.2	83.8	157.7	84.1	364	2.7	178.6	55.8	5.3	67.9	171.5	55.6	3.5
0306		12	7	150.8	85.6	184.4	84.6	136	5.2	169.2	56.2	10.2	74.2			
0308		12	8	167.4	85.3	194.1	83.7	696	2.1	165.5	54.7	4.1	66.4			
0309		12	7	186.9	77.1	195.9	75.1	189	4.4	161.1	39.4	7.7	28.6			
0310		15	8	213.4	82.7	220.7	79.9	213	3.8	150.8	50.1	7.1	25.3			
0305		15	14	127.3	87.1	184.5	86.7	123	3.6	169.7	60.3	7.1	41.8	172.8	58.2	5.1
0301		10	8	140.2	82.6	163.2	82.5	266	3.4	177.7	52.5	6.6	14.3			
Mean		**10**	**10**	**150**	**86.8**	**192.6**	**85.6**	**180**	**3.6**	**167.2**	**58.5**	**7**				
NRO										0.988				0.6276		

Directions

	n (*N*)	Dg	Ig	Kg	a95	Ds	Is	Ks	a95	
Site mean	74 (79)	350.8	82.7	40	2.6	4.7	84.9	44.8	2.5	
Mean over individual directions and directional groups	55(55)	352.6	83.1	37.4	3.2	6.8	85.2	40.8	3	
Section mean	9 (9)	338.7	82	117	4.8	356.3	85	176.4	3.9	

Poles

	n (*N*)	PLONG	PLAT	A95	Paleolatitude for conditional average point 66.9° N, 170.1° E
Site-mean pole	74 (79)	173.5	76.3	4.9	80.5 ± 4.9°
Mean pole over individual directions and directional groups	55 (55)	174.3	75.7	5.6	81.1 ± 5.6°
Section mean	9(9)	167.7	76.3	7.3	80.6 ± 7.3°

N is the number of samples taken from a site; *n* is the number of samples used in the calculation of paleomagnetic directions; Dg (Ds), Ig (Is) are inclinations and declinations in geographic (stratigraphic) coordinates; K, a95 are precision parameter and radius of confidence circle; Plon, Plat, A95 are the longitude and latitude of virtual geomagnetic poles (VGPs) for individual sites and groups of sites, respectively; Km is volumetric magnetic susceptibility; NRO is the non-random ordering parameter value.

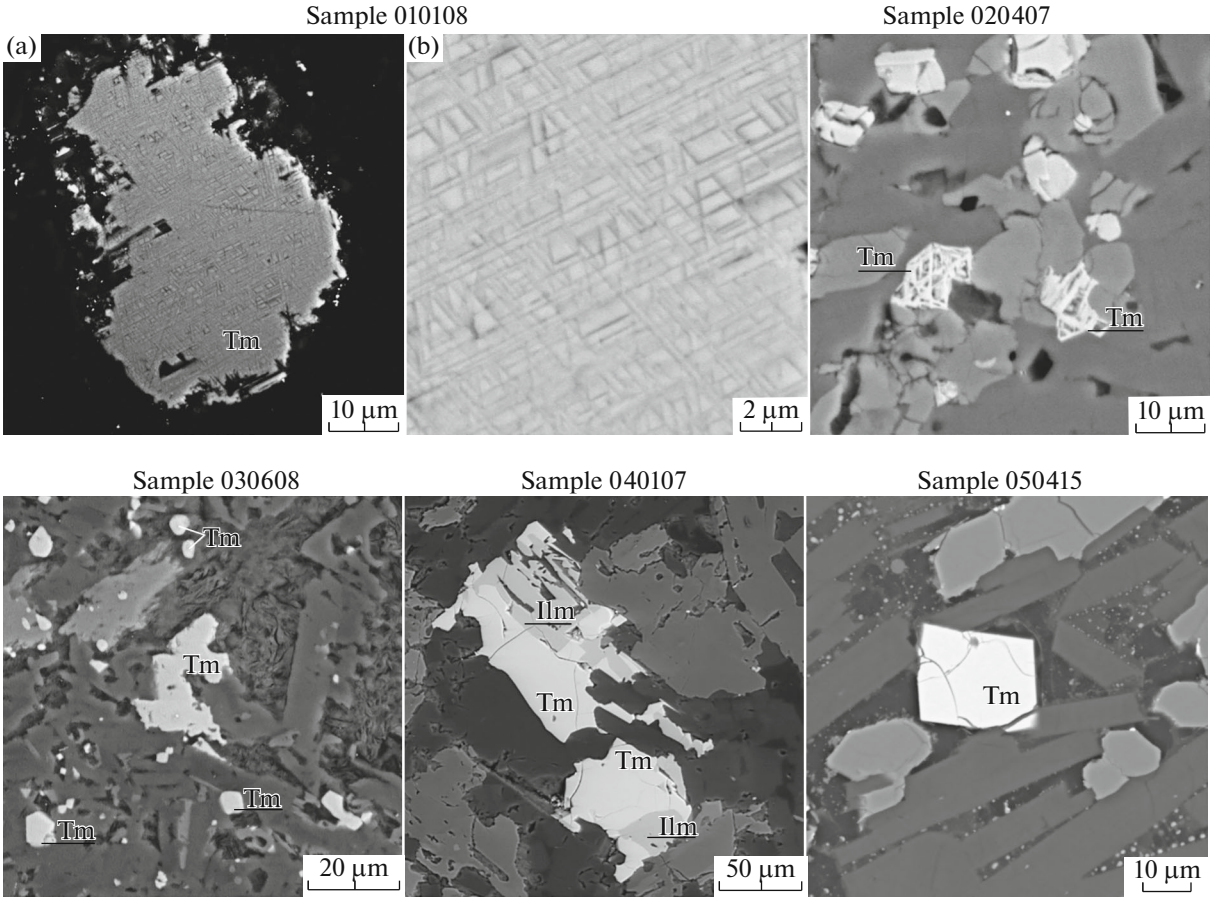


Fig. 4. Backscattered electron (BSE) images of polished thin sections, samples 010108 (section 01), 020407 (section 02), 030608 (section 03), 040107 (section 04): Tm, titanomagnetite; Ilm, ilmenite; Mt, magnetite.

nomagnetite grains (Fig. 5, samples 030608, 050415) and intergrowths of ilmenite (Ilm) with titanomagnetite (Tm) (Fig. 4, sample 040107). The presence of primary magmatic grains of magnetic minerals is a strong evidence in favor of the preservation of primary magnetization in the rocks.

In the scope of this work, we have also carried out some thermomagnetic and coercitimetric studies. As the obtained data do not provide additional information on the acquisition time of the characteristic magnetization, we decided to not discuss them in the paper.

Based on the results of our studies, we calculated the mean paleomagnetic pole (Table 1) whose coordinates are 75.7° N, 174.3° E at $N = 55$ and a radius of the confidence circle A95 = 5.6° (the calculation takes into account the directional groups, the details are in the next section).

The paleomagnetic pole of a similar age was previously obtained by Stone et al. (2009) from the OCVB volcanics outcropping close to our study region near Lake Elgygytgyn and in the upper reaches of the Enmyvaam River. The coordinates of this pole are 67.0° N, 171.0° E at A95 = 9.8° and $N = 40$ (Stone et al., 2009). The angular difference between these poles is insignificant: $-7.7 \pm 8.2°$ (Debiche and Watson, 1995), which again supports the reliability of our results.

DATA PROCESSING PROCEDURE A ND DISCUSSION OF THE RESULTS

The paleomagnetic data derived from the studied volcanics allow us to estimate the amplitude of high-latitude paleosecular geomagnetic variation during CNS on a time level of ~90 Ma ago.

Traditionally, the amplitude of paleosecular geomagnetic variation is described by a measure of scatter of the virtual geomagnetic poles (VGP) (Khramov et al., 1982) calculated for individual paleomagnetic sites. The VGPs in this case are determined by the dipole formula (Khramov et al., 1982) from the mean direction estimated for each studied flow. For the last 5 Ma, this scatter is considered with respect to the geographic pole. For more ancient time intervals, the VGP scatter is calculated either about the mean pole over the entire set of the obtained virtual poles or about the geographic pole but provided that the VGP

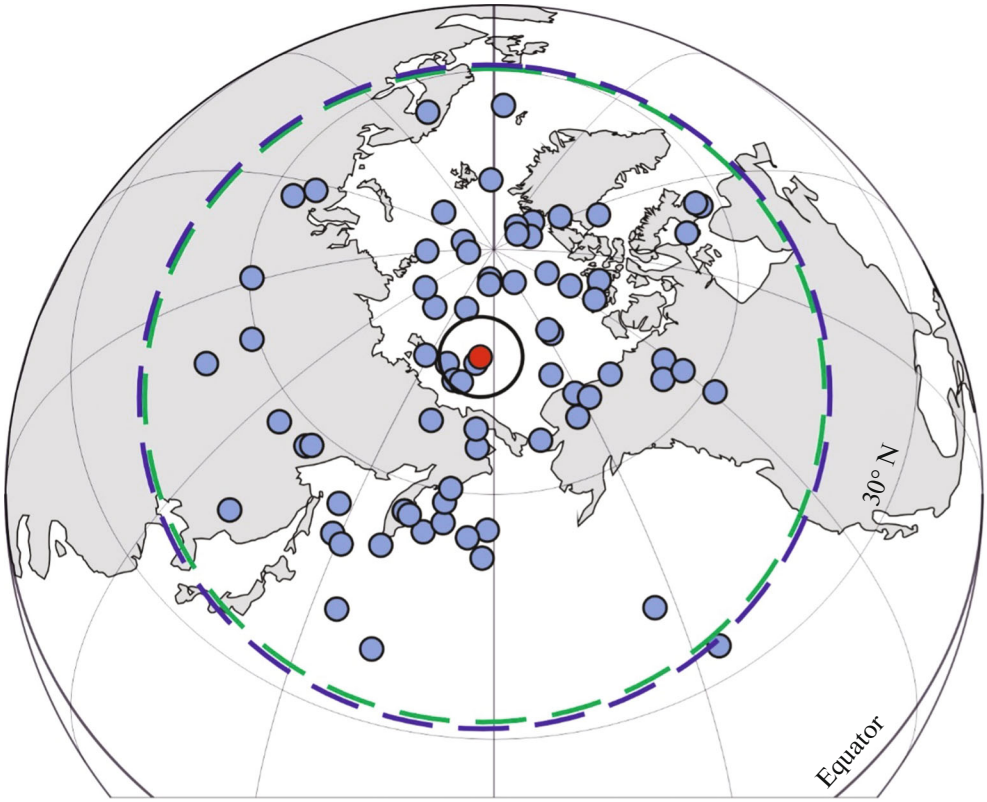


Fig. 5. Distribution of VGPs (purple circles) calculated over all 74 individual sites (flows). Red circle is mean paleomagnetic pole and radius of its 95% confidence circle; blue dashed circle is critical cutoff angle of 45°; green circle is critical cutoff angle (44.1°) calculated by Vandamme method.

position is corrected for the tectonic plate motion (Doubrovine et al., 2019). The VGP scatter is commonly estimated based on quantity *Sb* which is equal to RMS deviation of VGP from the geographic or from the mean pole corrected for the scatter of paleomagnetic directions within sites (Cox, 1970):

$$Sb = \left[\frac{1}{N-1}\sum_{i=1}^{N}\left(\Delta_i^2 - \frac{S_{wi}^2}{n_i}\right)\right]^{1/2} \quad (i = 1,\ldots,N), \quad (1)$$

where Δ_i is the angular distance of the ith VGP calculated for each ith flow from the mean VGP (or from the geographical pole); N is the number of flows and n_i is the number of samples taken from the ith flow; S_{wi} is the VGP dispersion within the site calculated for each ith flow by the following formula

$$S_{wi}^2 = 81\sqrt{K},$$

$$K = k\left(\frac{1}{8}(5 + 18\sin^2\lambda + 9\sin^2\lambda)\right)^{-1},$$

where k is the precision parameter of paleomagnetic directions within sampling sites and λ is the paleolatitude of the site (Cox, 1970).

VGP angular dispersion given by Eq. (1) includes two sources of dispersion: secular field variation (Δ_i^2) and random within-site scatter of paleomagnetic directions (S_{wi}). The first source provides the dominant contribution to the dispersion of the virtual geomagnetic poles.

We are interested in the PSV amplitude during "normal" (non-excursing, non-reversing) field states. Therefore, we have to discard the directions probably related to the time intervals when the field was in the excursing or reversing state. Besides, the crude measurement errors leading to the noticeable VGP deviations from the mean should also be eliminated. Since VGP deviations enter the *Sb* calculation quadratically, the VGPs associated with the "anomalous" field states or with crude errors will strongly affect the calculated *Sb* values.

Such VGPs are rejected based on some threshold value—a critical *cutoff* or a *cutoff angle*. Virtual poles that deviate from the mean pole by an angle greater than the cutoff angle are rejected, and only the VGPs that are distant from the mean pole by a smaller angle are included in the *Sb* calculation. The critical angle (Δmax) is either selected constant, e.g., 45° (a review is presented in (McElhinny and McFadden, 1997)) or calculated using the Vandamme's iterative method (Vandamme, 1994):

$$\Delta\max = 1.95S' + 5°,$$

where S' is equal to *Sb* for the initial and each subsequent distribution obtained with a current rejection of anomalous VGPs.

Following (Biggin et al., 2008; Pavlov et al., 2019; Doubrovine et al., 2019), we used two approaches to remove the VGP outliers. According to the first method, the critical angle was selected fixed and equal to 45°; according to the second method, this angle was calculated based on the Vandamme method (Fig. 5).

Observations in the regions of the present-day continental margin volcanism show that volcanic flows are formed irregularly. Single volcanic eruptions separated by fairly long time intervals can produce single flows. Volcanic pulses consisting of a series of eruptions can form a series (groups) of practically coeval or close in age volcanic flows. In the latter case, in the *Sb* calculations at the level of individual flows, the contribution of the field states during which these volcanic series were formed will be greater than the contribution of the field states recorded in the single flows, which will result in the underestimation of the calculated *Sb* values relative to the true value. To avoid this, we used two procedures: comparing the mean paleomagnetic directions of neighboring flows by the method (McFadden and McElhinny, 1991) and estimating the serial correlation of paleomagnetic directions within the studied sections by the method described in (Biggin et al., 2008).

Using the first procedure, we identified the groups of the successive flows with statistically indistinguishable mean paleomagnetic directions. Accordingly, the entire data for calculating the variation were represented as a set of individual directions corresponding to the individual flows and directional groups corresponding to the groups (series) of the coeval or almost coeval flows (see Table 1). As a result, a total of 55 directions (VGPs) determined from 43 individual flows (sites) and 12 groups were accepted for the final calculation.

The second procedure calculates the NRO (nonrandom ordering) parameter allows estimation of the degree of serial correlation of paleomagnetic directions in a section (Biggin et al., 2008). NRO above 0.95 means that the correlation of paleomagnetic directions in a section (serial correlation) is significant at 95% confidence level and the obtained sequence of directions probably reflects the close-in-time states of the field. The latter may lead to an underestimation of the PSV amplitude. If NRO is below 0.95, it can be assumed that the considered paleomagnetic directions are uncorrelated, reflect independent states of the geomagnetic field states, and can be confidently used for calculating the unbiased estimate of the amplitude of the paleosecular variation.

We calculated the NRO parameter for each studied section (Table 1). The calculations were carried out both for the single flows and for the sequences of directions representing individual directions and directional groups.

In four sections (1, 2, 3, and A1) of the nine studied, the NRO value before grouping the flows was above 0.95 (Table 1). After the grouping, NRO dropped notably below this threshold value (Table 1). Hence, the directions obtained after grouping form a statistically random field sample and fairly adequately represent the dimension of the paleosecular variation.

This result provides us with an additional indication that the *Sb* calculation from directional groups and individual directions is more correct than the calculation at the level of individual sites without grouping the directions.

Nevertheless, for comparison, we calculated *Sb* both with and without grouping (Table 2). Calculations were conducted with the fixed cutoff angle *cutoff* = 45° and with the cutoff angle calculated by the Vandamme method. To assess how the quality of the paleomagnetic record affects the final result of *Sb* determination, we performed calculations using all individual directions and groups and using only those where the within-site (within-group) precision parameter K is greater than 50 and 100. The results are presented in Table 2 and Fig. 6.

From Table 2 and Fig. 6 it can be seen that irrespective of the calculation method, the *Sb* values fall in a narrow range from 21.3° to 23.5°, and the confidence intervals for these quantities substantially overlap making them statistically indistinguishable. We note, however, that in the case of using the sites (groups) with $K > 100$, the *Sb* value always but insignificantly increases and the confidence interval increses. For our analysis, we select the *Sb* values obtained with grouping the directions for the sites and the groups where the precision parameter is higher than 50. Considering the fact that in the paleosecular variation studies, both fixed cutoff angles and the cutoff angles calculated by the Vandamme method are widely used, below we will analyze both the corresponding *Sb* values (Table 2): 22.1° [19.5°; 24.7°] and 23.1° [20.3°; 25.7°].

Let us compare the obtained values with the models of PSV latitudinal dependence for CNS proposed in (McFadden et al., 1991; Biggin et al., 2008; Doubrovine et al., 2019). As can be seen from Fig. 7a, when we use a fixed cutoff angle, the resulting *Sb* values excellently agree with the model of (Doubrovine et al., 2019) and slightly differ from the values predicted by the model of (McFadden et al., 1991). In the case of the cutoff angle determined by the Vandamme method (Fig. 7b), our result nonsignificantly differs from the model (Doubrovine et al., 2019) but even better agrees with the model (Biggin et al., 2008).

Thus, our data for the high latitudes generally support the existing models. The agreement with the model (McFaddenetal., 1991) where the PSV values are low at the equator and sharply increase latitude is worse than with the model (Biggin et al., 2008) which

Table 2. PSV amplitude estimates

Selection criteria	*n*(*N*)	*Sb* [*Sb*–; *Sb*+]
All sites without grouping, fixed cutoff angle *cutoff* = 45°		
All	73(74)	*21.3 [19.1; 23.5]*
K > 50	72(73)	21.2 [19.0; 23.4]
K > 100	55(56)	22.0 [19.4; 24.4]
All sites without grouping, cutoff angle determined by Vandamme method		
All, *cutoff* = 44.1°	73(74)	*21.7 [19.5; 23.9]*
K > 50, *cutoff* = 44.0°	72(73)	21.7 [19.3; 23.8]
K > 100, *cutoff* = 46.2°	56(56)	22.9 [20.1; 25.7]
Sites with grouping, fixed cutoff angle *cutoff* = 45°		
All	54(55)	*22.1 [19.0; 24.6]*
K > 50	53(54)	*22.1 [19.5; 24.7],*
K > 100	41(42)	22.4 [19.1; 25.6]
Sites with grouping, cutoff angle determined by Vandamme method		
All, *cutoff* = 46.7°	55(55)	23.2 [20.2; 26.0]
K > 50, *cutoff* = 46.6°	54(54)	23.1 [20.3; 25.7]
K > 100, *cutoff* = 47.3°	42(42)	23.5 [20.2; 26.7]

N is the number of sites (individual flows, individual directions) or number of individual flows and flow groups (individual directions and directional groups); *n* is the number of sites (individual flows, individual directions) or number of individual flows and flow groups (individual directions and directional groups) used in *Sb* calculation after cutting off at critical cutoff angle; *cutoff* is the critical cutoff angle to eliminate VGP "outliers"; *K* is the within-site (within-group) precision parameter of paleomagnetic directions. *Sb*° is the VGP dispersion (amplitude of variations); *Sb*+ and *Sb*− are the upper and lower limits of the 95% confidence interval.

predicts higher PSV values at the equator and their more gradual increase towards polar latitudes. We recall that it is these PSV properties that have been considered until recently as most promising for revealing the differences in the behavior of the geomagnetic field during and between the geomagnetic

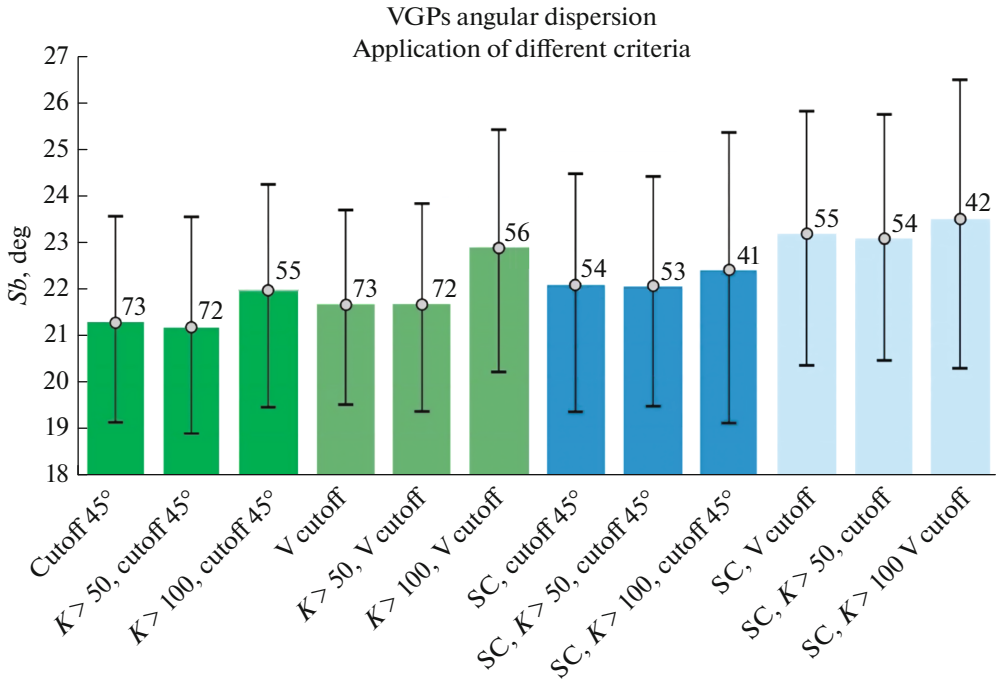


Fig. 6. *Sb*° values (VGPs angular dispersion) and their 95% confidence intervals for OCVB volcanic flows calculated with different flow selection criteria. Numerals at top of bars are number (*N*) of flows or DG&DI after rejection by cutoff angle and precision parameter (*K*). Different criteria and cutoffs are indicated on *X*-axis, where V is iterative cutoff according to (Vandamme, 1994), 45° is fixed cutoff, and SC means application of serial correlation procedure (blue bars).

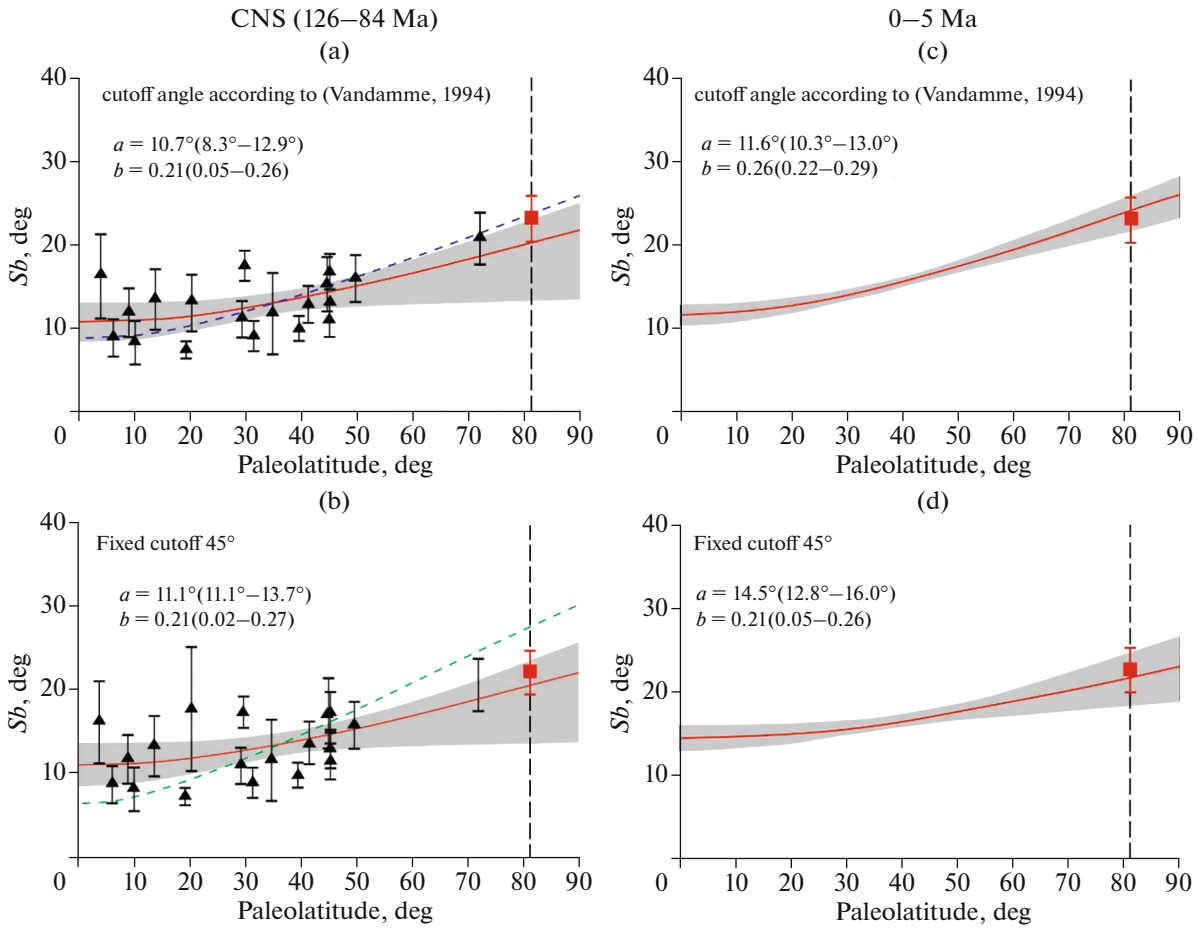


Fig. 7. Based on (Doubrovine et al., 2019) with modifications and additions. (a), (b) Comparison of obtained data with models of PSV latitudinal dependence during CNS (84–126 Ma): (a) with fixed cutoff angle 45°; (b) with cutoff angle calculated by Vandamme method; (c), (d) comparison of obtained data with models of PSV latitudinal dependence for last 5 Ma: (c) with cutoff angle calculated by Vandamme method (Opdyke et al. , 2015); (d) with fixed cutoff angle 45° (Johnson et al., 2008). Red square and corresponding line show *Sb* values and their confidence intervals obtained in this study. Red solid line is data approximation in (Doubrovine et al., 2019) based on model (McFadden et al., 1988) with parameters *a* and *b* and their confidence intervals (in parentheses) shown on diagram. Gray shading shows 95% confidence intervals for corresponding models. Black triangles and corresponding segments show data of other authors for CNS. Blue (green) dashed line shows approximation of latitude dependence of PSV according to (Biggin et al., 2008; McFadden et al., 1991).

superchrons. Our results do not support this standpoint.

It is also interesting to compare our data for the superchron with the results for the last five million years when the rate of geomagnetic reversals was quite high averaging about five reversals per million years.

From Figs. 7c and 7d it can be seen that our results almost perfectly fit the latitude dependences of PSV for the last 5 Ma irrespective of whether the calculations used a fixed cutoff angle (Johnson et al., 2008) or an angle determined by the Vandamme method (Opdyke et al., 2015). This means that at least at high latitudes, the amplitude of secular geomagnetic variation during CNS was the same as during the last 5 Ma.

The obtained results may have important implications for the discussion of the fundamental question of whether the superchrons reflect a special physical state of the geodynamo or are the manifestation of the same stochastic process that is responsible for the "normal reversing regime." Our data rather support the second standpoint.

If this standpoint is right, we can speak of incomplete adequacy of the modern numerical geodynamo models for the real processes of geomagnetic field generation. This is due to the fact that all these models require a change in the geodynamo operation conditions during superchrons (e.g., a change in the convection parameters due to the change in the magnitude or character of distribution of the heat flow at CMB).

The absence of clear differences in the amplitude of secular variation during the Cretaceous superchron and during the last 5 Ma does not support the hypothesis that the regime of geomagnetic field generation changes during the establishment and termination of

the superchrons (e.g., (Gallet and Pavlov, 2016)). Otherwise, it would have to be assumed that the change of the generation regimes does not necessarily lead to the substantial changes in the PSV amplitude, which raises new questions for the modern geodynamo models.

Speaking about the absence of a sharp contrast between the amplitude of secular variation during the Cretaceous superchron and that during the last five million years, we have to make one important reservation. Recently, the data have emerged probably indicating that the scatter of paleomagnetic directions could markedly increase on some time intervals within CNS (Guzhikov and Feduleev, 2019; Guzhikova et al., 2020; Lhuillier et al., 2016). Whether this increase is related to the widening of the scatter around a single central direction or to the presence of several (two?) clusters of the preferred VGP positions is unclear. It is obvious that the solution of these questions requires further accumulation of the new high-quality data on the secular variation in the geomagnetic field of the superchrons and periods of frequent reversals.

CONCLUSIONS

The paleomagnetic studies of the Okhotsk–Chukotka belt volcanics outcropping in the region of Kupol deposit (~66.9° N, 170.1° E) and formed about 90 Ma ago during CNS have been carried out. The characteristic magnetization component is isolated in most of the studied samples. The primary origin of this component is supported by the positive fold test, the electron microscopic and microprobe analysis data, and the closeness of the calculated paleomagnetic pole to that of the coeval rocks from the same region. It is shown that the studied rocks were formed at high polar latitudes close to 81° N.

The amplitudes of the geomagnetic paleosecular variation (PSV) during CNS have been estimated. Depending on the calculation method, these estimates for ~81° N (and their confidence intervals) are Sb = 22.1° (19.5°; 24.7°) when calculated with a fixed cutoff angle of 45° and Sb = 23.18° (20.3°; 25.7°) when calculated with the cutoff angle determined by the Vanndamme method.

The obtained estimates generally support the existing models of the latitudinal dependence of PSV for CNS. The agreement with the model (McFadden et al., 1991) in which PSV is assumed to be low at the equator and to sharply increase with latitude is worse than with the model (Biggin et al., 2008) which predicts higher PSV values at the equator and their more gentle increase towards polar latitudes.

Irrespective of the calculation method, the obtained estimates almost perfectly fit the latitude dependences of PSV for the last 5 Ma. This means that the amplitude of secular geomagnetic variation during CNS, at least at high latitudes and during the time interval under study, was the same as in the last 5 Ma.

The presented result does not support the hypothesis of a change in the geodynamo regime at the transitions from the periods of frequent geomagnetic reversals to superchrons, which suggests incomplete adequacy of the modern numerical models for the real processes of generation of the geomagnetic field.

ACKNOWLEDGMENTS

We are particularly grateful to V.A. Tselmovich and N.N. Koshlyakova for their help in conducting the electron microscopic studies and microprobe analysis. All measurements have been carried out at the IPE RAS Research Equipment Sharing Center "Petrophysics, geomechanics, and paleomagnetism."

FUNDING

The work was supported by the Russian Science Foundation under project no. 19-47-04110 (field work, demagnetization experiments, data interpretation) and carried out in partial fulfillment of the state contract of the North-Eastern Interdisciplinary Scientific Research Institute, Far Eastern Branch RAS (Magadan) under project no. 121031700312-1 (petrographic studies).

CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

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Translated by M. Nazarenko